

Pharmacology 1: Receptor Mechanisms Examination

Part I: Multiple Choice Questions (MCQs)

Section 1: Basic Definitions (Receptors, Affinity, Activity)

1. What are macromolecules located on the cell surface, cytoplasm, or nucleus that bind with drugs to produce cellular changes?
 - A. Enzymes
 - B. Receptors
 - C. Transporters
 - D. Hormones
2. The ability of a drug to bind specifically to a receptor is known as:
 - A. Intrinsic activity
 - B. Efficacy
 - C. Affinity
 - D. Potency
3. "Intrinsic activity" is best defined as the ability of a drug to:
 - A. Bind to the receptor
 - B. Produce pharmacological action after binding
 - C. Be excreted from the body
 - D. Cross the cell membrane

Section 2: Agonists, Antagonists, and Other Ligands

4. A drug that possesses both high affinity and high intrinsic activity is called a/an:
 - A. Antagonist
 - B. Partial Agonist
 - C. Agonist
 - D. Inverse Agonist
5. Which of the following is a classic example of an agonist mentioned in the sources?

- A. Atropine
 - B. Adrenaline
 - C. Propranolol
 - D. Salbutamol
6. An antagonist is a drug that:
- A. Has high affinity and high intrinsic activity.
 - B. Has high affinity but NO intrinsic activity.
 - C. Produces the opposite effect of the natural ligand.
 - D. Has low affinity and low activity.
7. A drug like Atropine, which binds to a receptor but produces no pharmacological action (only a blockade), is a:
- A. Full Agonist
 - B. Competitive Antagonist
 - C. Partial Agonist
 - D. Inverse Agonist
8. A "Partial Agonist" is characterized by:
- A. High affinity and no activity.
 - B. Affinity and an effect less than that of a full agonist.
 - C. Producing the opposite effect of an agonist.
 - D. Binding irreversibly to the receptor.
9. Which type of drug has an intrinsic activity between 0 and -1?
- A. Agonist
 - B. Antagonist
 - C. Partial Agonist
 - D. Inverse Agonist

Section 3: Receptor Families

10. Receptors that are directly coupled to ion channels and respond in milliseconds are:

- A. G-protein-coupled receptors
- B. Ligand-gated ion channels (Inotropic)
- C. Enzymatic receptors
- D. Nuclear receptors

11. G-protein-coupled receptors (metabotropic) are linked to effectors through G-proteins, which consist of how many subunits?

- A. Two (α , β)
- B. Three (α , β , γ)
- C. Four (α , β , γ , δ)
- D. One (α)

12. The insulin receptor is an example of which receptor family?

- A. Ligand-gated ion channel
- B. G-protein-coupled receptor
- C. Enzymatic receptor (Tyrosine kinase)
- D. Transcription factor

13. Which receptor family is located intracellularly and regulates gene expression?

- A. GABA receptors
- B. Muscarinic receptors
- C. Adrenergic receptors
- D. Nuclear receptors (e.g., Steroids)

14. How long does it typically take for Nuclear receptors to produce a response?

- A. Milliseconds
- B. Seconds
- C. Hours
- D. Minutes

Section 4: Regulation of Receptors

15. Prolonged use of an agonist (like Salbutamol) often leads to:
- A. Down-regulation (decreased receptor number and sensitivity)
 - B. Up-regulation (increased receptor number)
 - C. No change in receptors
 - D. Supersensitivity
16. Sudden stoppage of an antagonist (like Propranolol) after prolonged use can cause:
- A. Down-regulation
 - B. Up-regulation/Supersensitivity (withdrawal symptoms)
 - C. Permanent receptor blockade
 - D. Decreased response to catecholamines
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Part II: Essay Questions

1. Compare Agonists, Antagonists, Partial Agonists, and Inverse Agonists based on their affinity and intrinsic activity.
 2. Tabulate the four receptor families, including their location, coupling mechanism, and one example for each.
 3. Discuss the clinical significance of "Up-regulation" of receptors, using Propranolol as an example.
 4. Explain the mechanism of "Down-regulation" and how it affects the treatment of asthmatic patients using Salbutamol.
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Answer Key

Part I: MCQs

1. B

2. C
3. B
4. C
5. B
6. B
7. B
8. B
9. D
10. B
11. B
12. C
13. D
14. C
15. A
16. B

Part II: Essay Hints

1. Agonist: High affinity + High intrinsic activity (+1). Antagonist: High affinity + Zero activity (0). Partial Agonist: Affinity + Less activity (<1). Inverse Agonist: Affinity + Opposite activity (0 to -1).
2. Ion Channels: Membrane, Direct coupling, e.g., Nicotinic. G-Protein: Membrane, G-proteins, e.g., Adrenergic. Enzymatic: Membrane, Direct enzyme, e.g., Insulin. Nuclear: Intracellular, DNA/Gene, e.g., Steroids.
3. Up-regulation: Chronic antagonist use increases receptor number/sensitivity. Sudden stoppage leads to a massive response to natural agonists (e.g., catecholamines), causing tachycardia or angina in patients stopping Propranolol.
4. Down-regulation: Chronic agonist use (Salbutamol) decreases receptor number/sensitivity, leading to a decreased drug effect in asthmatic patients.